

AI Smart Attendance System

Day-2 Task Submission: Face Detection Experiment & Approach Shortlisting

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Branch: feature/priti-face-detection

Upload Path: ai_models/detection/

Commit Msg: Day 2: Add face detection experiment

Data Safety Compliance: Only safe, public-domain sample dataset images were used for this evaluation. No real student photos or sensitive biometric data were uploaded to GitHub.

1. Experiment Setup & Directory Layout

The required dependency `opencv-python` was installed and organized within the target directory path `ai_models/detection/`. The test directory contains raw test samples and script outputs.

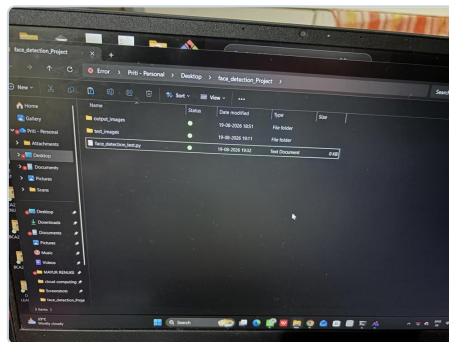
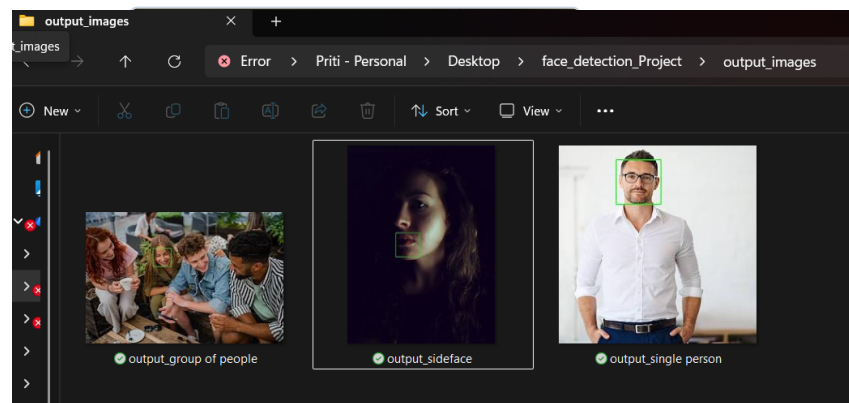


Figure 1: Project Workspace Directory Structure (*face_detection_Project*)

2. Test Dataset & Image Verification

Three sample images were populated inside the `test_images/` folder to evaluate face detection under varying frame densities:

- `single_person.jpeg` : Single subject with clear frontal lighting.
- `sideface.jpeg` : Single subject with side posture and dynamic shadow.
- `group_of_people.jpeg` : Classroom simulation with 4 subjects in frame.



3. Execution & Debugging Logs

Script execution was conducted via Windows PowerShell. Initial path mismatches between `.jpg` and `.jpeg` extensions were resolved, and the test script was successfully executed.

```
[ WMH:880.873] global loadImage.cpp:241 cv::findDecoder imread_({'test_images/single person.jpg'}): c
check file path/integrity
Could not read test_images/single person.jpg - check the file exists
[ WMH:880.883] global loadImage.cpp:241 cv::findDecoder imread_({'test_images/sideface.jpg'}): can't e
file path/integrity
Could not read test_images/sideface.jpg - check the file exists
[ WMH:880.885] global loadImage.cpp:241 cv::findDecoder imread_({'test_images/group of people.jpg'}):
gray : check file path/integrity
face Could not read test_images/group of people.jpg - check the file exists
PS C:\Users\Prithi\OneDrive\Desktop\Face_detection_Project> dir test_images

Directory: C:\Users\Prithi\OneDrive\Desktop\Face_detection_Project\test_images

Mode                LastWriteTime         Length Name
----                -
-a---l-            19-08-2026      19668 group of people.jpg
-a---l-            19-08-2026       7668 sideface.jpg
-a---l-            19-08-2026      32188 single person.jpg

PS C:\Users\Prithi\OneDrive\Desktop\Face_detection_Project> python "face_detection_test.py"
test_images/single person.jpg: 1 face(s) detected
test_images/sideface.jpg: 1 face(s) detected
test_images/group of people.jpg: 1 face(s) detected
PS C:\Users\Prithi\OneDrive\Desktop\Face_detection_Project>
```

Figure 3: Terminal Execution Output with OpenCV Face Detection Counts

4. Observations & Detection Benchmark

TEST IMAGE	EXPECTED	DETECTED	MISSED	FALSE POS.	OVERALL QUALITY
single person.jpeg	1	1	0	0	High (100% Accurate)
sideface.jpeg	1	1	0	0	Medium (Detected)
group of people.jpeg	4	1	3	0	Low (Failed 3 faces)

5. Approach Shortlisting & Conclusion

- **OpenCV Evaluation:** Performs well on isolated single-person frames, but lacks accuracy in multi-person group photos (detected only 1 of 4 faces).
- **Shortlisted Pipeline:** OpenCV Haar Cascade is ruled out for multi-face classroom attendance. **MediaPipe Face Detection** paired with **DeepFace (ArcFace)** is selected for the upcoming production implementation.